Lecture 25
Composite hypotheses

Review

Testing a mean: σ known

when to use it

σ is known; a random sample from a normal population, or n ≥ 30

test statistic

$z_{obs} = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$

if H_0 is true

z_{obs} ~ N(0, 1), standard normal

P-value

from the z-table, e.g. $P(Z \geq z_{obs} \mid H_0 \text{ true})$ if upper-tailed

Testing a mean: σ unknown

when to use it

σ is unknown, so we estimate it with the sample SD s

test statistic

$t_{obs} = (\bar{x} - \mu_0) / (s/\sqrt{n})$, df = n – 1

if H_0 is true

t_{obs} ~ t(n – 1), Student's t: heavier tails than N(0, 1)

P-value from the t-table

find t_{obs} between two values in row df; P is between their tail areas

Example: Marco's throwing distance

Marco, a high school quarterback, throws a football a mean of 40 yards, with σ = 2. His coach has him adjust his grip and records 20 throws. Distances are normal. (OpenStax, Try It 9.14)

$n = 20$; $\bar{x} = 45$ yards; $\sigma = 2$; $\mu_0 = 40$; $\alpha = 0.05$

(a) What are H_0 and H_A? Which tail? z or t?
(b) What are the standard error and the test statistic?
(c) What is the P-value? What is the decision?
(d) What is the critical value? Does z_{obs} fall in the rejection region?

(a) $H_0: \mu = 40$ $H_A: \mu > 40$, upper-tailed; σ known, so z
(b) $SE = 2 / \sqrt{20} \approx 0.4472$; $z_{obs} = (45 - 40) / 0.4472 \approx 11.18$
(c) $P(Z \geq 11.18 \mid H_0 \text{ true}) \approx 0 < 0.05$: reject H_0
(d) $Z_{0.95} = 1.645$; $z_{obs} = 11.18 \geq 1.645$, in the rejection region: reject H_0

Example: the first statistics test

Students believe the mean score on the first statistics test is 65. An instructor thinks it is higher and samples 10 students' scores, which are normal. (OpenStax, Example 9.16)

$n = 10$; $\bar{x} = 67$; $s = 3.197$; $\mu_0 = 65$; $\alpha = 0.05$

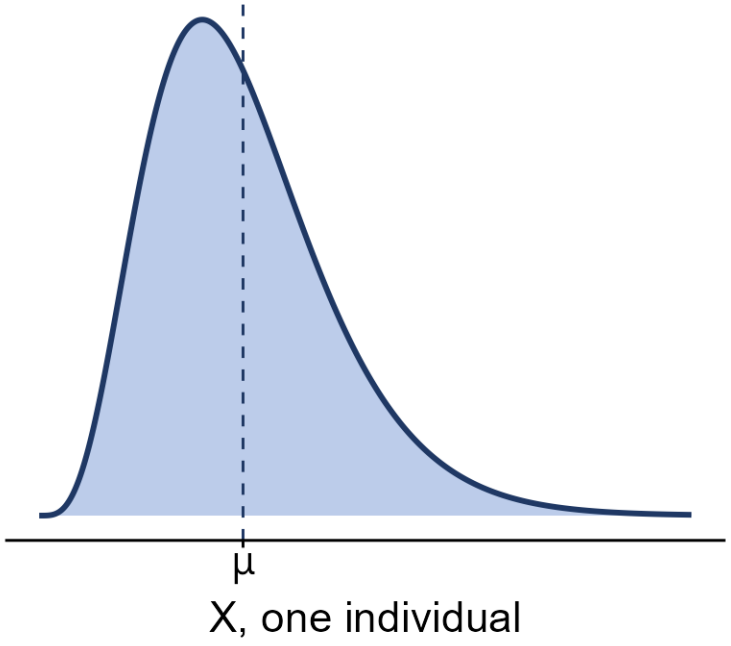
(a) What are H_0 and H_A? Which tail? Are the assumptions met?
(b) z or t? What are the standard error and the test statistic?
(c) What is the P-value (use the t-table)? What is the decision?
(d) What is the critical value? Does t_{obs} fall in the rejection region?

(a) H_0: $\mu = 65$ H_A: $\mu > 65$, upper-tailed; $n < 30$, but the scores are normal
(b) σ unknown, so t; $SE = 3.197 / \sqrt{10} \approx 1.011$; $t_{obs} = (67 - 65) / 1.011 \approx 1.98$, $df = 9$
(c) row 9: $1.833 < 1.98 < 2.262$, so $0.025 <$ P-value < 0.05: reject H_0
(d) $t_{9, 0.95} = 1.833$; $t_{obs} = 1.98 \geq 1.833$, in the rejection region: reject H_0

Three distributions: a mean

Distribution of X

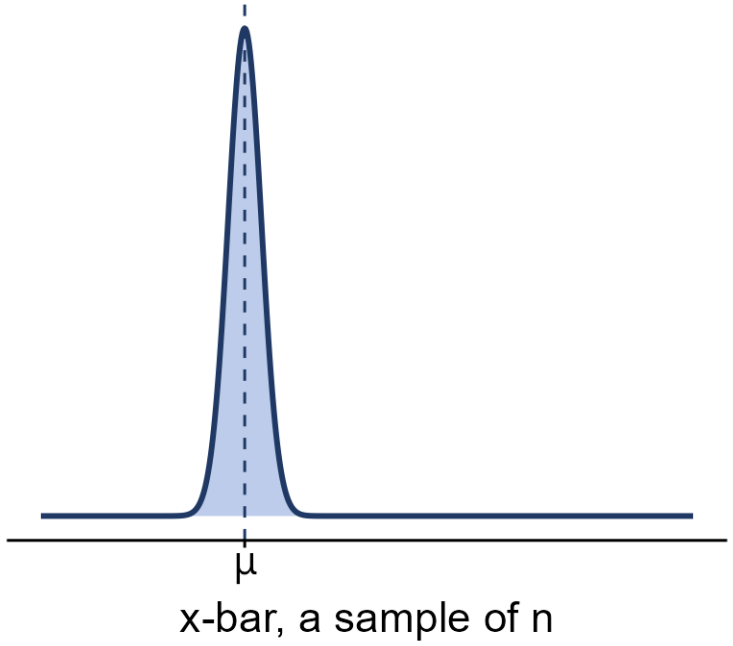
mean μ, SD σ, any shape



X, one individual

Sampling dist. of the mean

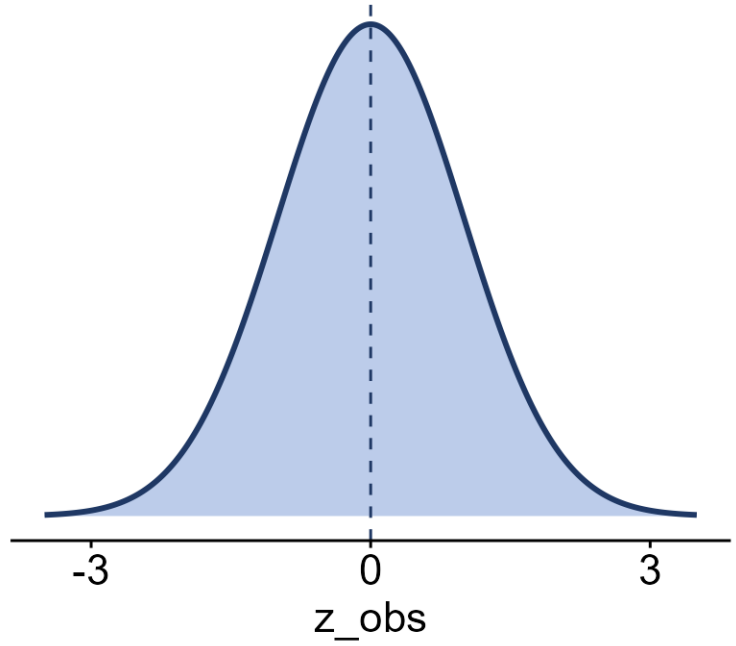
$\bar{x} \sim N(\mu, \sigma/\sqrt{n})$



x-bar, a sample of n

Distribution of z_obs

$z_{obs} \sim N(0, 1)$, if H_0 true

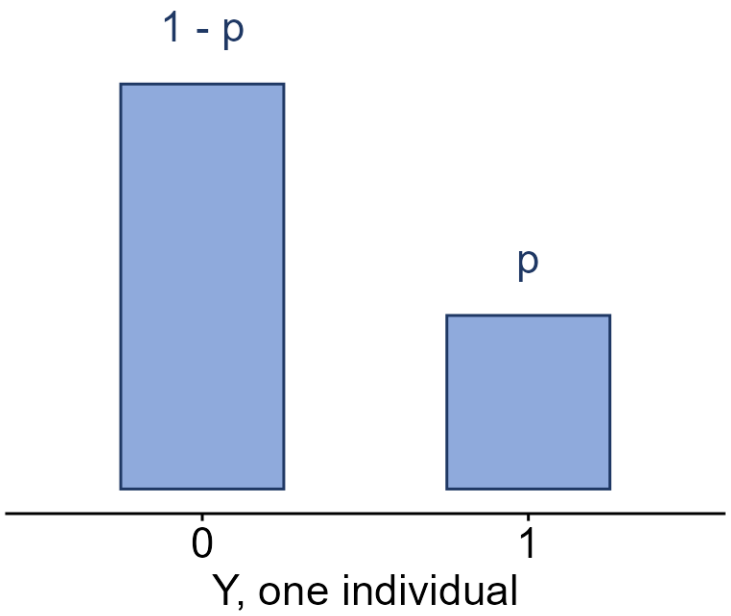


z_obs

Three distributions: a proportion

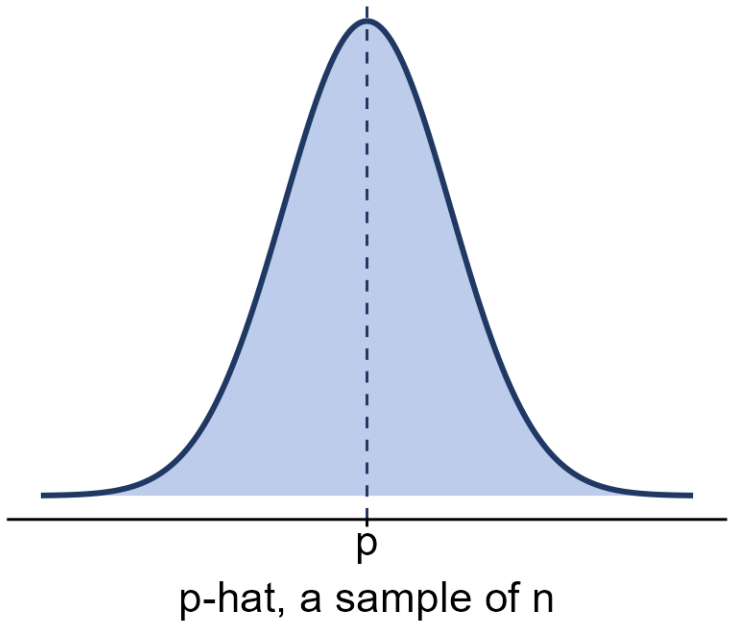
Distribution of Y

$Y \sim \text{Bernoulli}(p)$



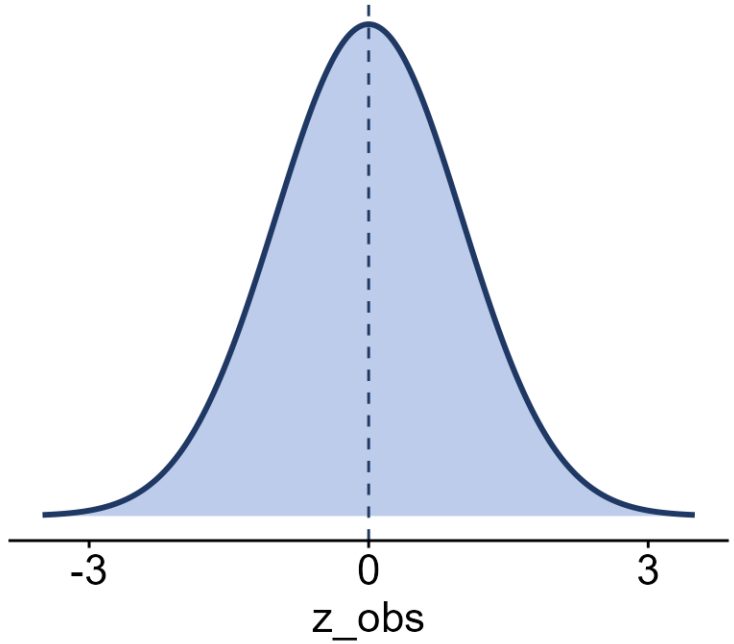
Sampling dist. of p-hat

$\hat{p} \sim N(p, \sqrt{p(1-p)/n})$



Distribution of z_obs

$z_{obs} \sim N(0, 1)$, if H_0 true



Why is z_{obs} standard normal?

a z-score

(value − mean of its distribution) / SD; ~ N(0, 1) if the value is normal

if H_0 is true

$\mu = \mu_0$, so $\bar{x} \sim N(\mu_0, \sigma/\sqrt{n})$

so z_{obs} is a z-score

$z_{obs} = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$ is the z-score of $\bar{x}$: ~ N(0, 1)

if H_0 is false

z_{obs} is centred at $(\mu - \mu_0) / (\sigma/\sqrt{n})$, away from 0

Example: the bench press

A coach's records say his players bench press a mean of 275 pounds, with σ = 55. Three players think it is more and ask 30 teammates. (OpenStax, Example 9.15)

$n = 30$; $\bar{x} = 286.17$; $\sigma = 55$; $\mu_0 = 275$; $\alpha = 0.025$

(a) If H_0: $\mu = 275$ is true, what is the distribution of $\bar{x}$?
(b) How many SDs above 275 is $\bar{x}$? What is z_{obs}?
(c) What is the P-value for H_A: $\mu > 275$? What is the decision?

(a) $\bar{x} \sim N(275, 55/\sqrt{30}) = N(275, 10.04)$, normal
(b) $(286.17 - 275) / 10.04 \approx 1.11$ SDs: $z_{obs} = 1.11$
(c) $P(Z \geq 1.11 \mid H_0 \text{ true}) = 1 - 0.8665 = 0.1335 \geq 0.025$: do not reject H_0

Composite hypotheses

simple hypothesis

names one value of the parameter: $H_0: \theta = \theta_0$

composite hypothesis

allows a range of values: $H_0: \theta \leq \theta_0$ or $H_0: \theta \geq \theta_0$

H_0 always has the equals sign

=, ≤ or ≥ (H_A never does); a two-sided test always uses $H_0: \theta = \theta_0$

test at the boundary

compute z_{obs} or t_{obs} with θ_0, the edge of H_0: the hardest value to reject

Practice: voter turnout

A county election official reports that no more than 30% of the registered voters in Santa Clara County voted in the primary election. A local newspaper suspects turnout was higher, and surveys a random sample of registered voters. (OpenStax, Example 9.1)

the official's figure: no more than 30% voted

(a) What is the parameter, in words?
(b) State H_0 and H_A.
(c) Is H_0 simple or composite? Which tail?

(a) p = the proportion of all registered voters in the county who voted in the primary
(b) H_0: $p \leq 0.30$ H_A: $p > 0.30$
(c) composite; upper-tailed, tested at the boundary $p_0 = 0.30$

Practice: time to graduate

College students are often told a degree takes five years, on average. An advising office believes its students finish faster, and records how long a random sample of recent graduates took to graduate. (OpenStax, Example 9.3)

the usual figure: 5 years, on average

(a) What is the parameter, in words?
(b) State H_0 and H_A.
(c) Is H_0 simple or composite? Which tail?

(a) μ = the mean time to graduate for all of the college's students
(b) H_0: $\mu \geq 5$ H_A: $\mu < 5$
(c) composite; lower-tailed, tested at the boundary $\mu_0 = 5$

Practice: a cure-rate claim

A drug company advertises that its experimental drug cures at least 75% of men with prostate cancer. An independent research group suspects the true cure rate is lower, and follows a random sample of patients who took the drug. (OpenStax, Example 9.8)

the company's claim: a cure rate of at least 75%

(a) What is the parameter, in words?
(b) State H_0 and H_A.
(c) Is H_0 simple or composite? Which tail?

(a) p = the proportion of all men with prostate cancer who are cured by the drug
(b) H_0: $p \geq 0.75$ H_A: $p < 0.75$
(c) composite; lower-tailed, tested at the boundary $p_0 = 0.75$

Example: the conductivity of glass

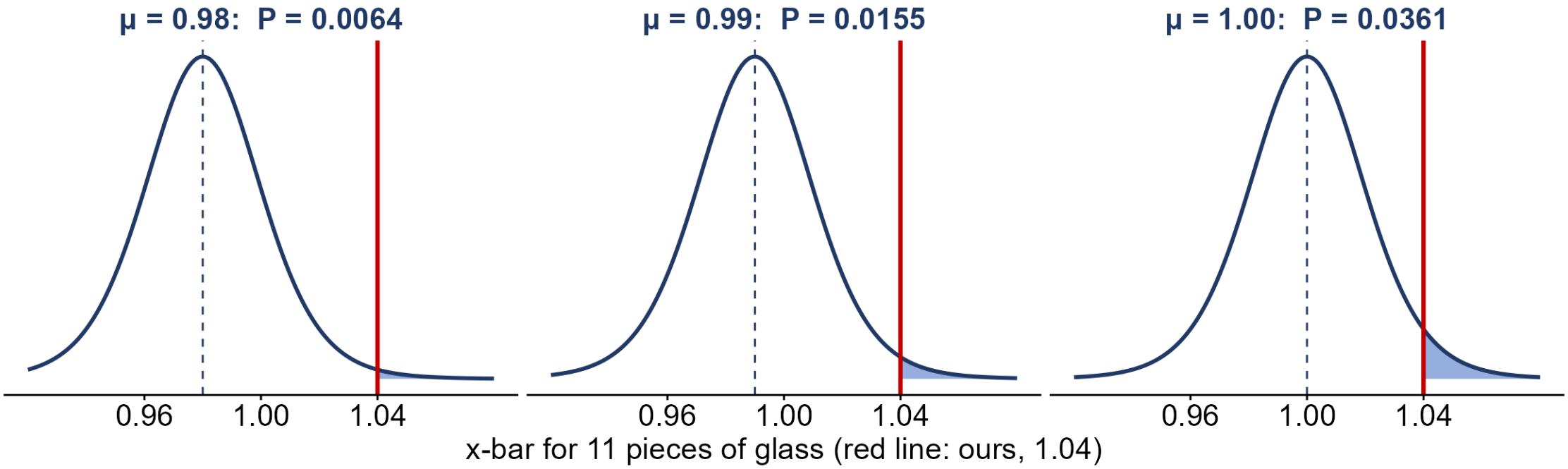
NIST measures the conductivity of 11 randomly selected pieces of a type of glass. Is the mean greater than 1? The population is normal. (OpenStax, Example 9.20)

$n = 11$; $\bar{x} = 1.04$; $s = 0.0659$; $\alpha = 0.05$

(a) What are H_0 and H_A? Which value of μ do we test at?
(b) What are the standard error and t_{obs}?
(c) What is the P-value (use the t-table)? What is the decision?

(a) H_0: $\mu \le 1$ (composite) H_A: $\mu > 1$; test at the boundary, $\mu_0 = 1$
(b) SE = $0.0659 / \sqrt{11} \approx 0.0199$; $t_{obs} = (1.04 - 1) / 0.0199 \approx 2.01$, df = 10
(c) row 10: $1.812 < 2.01 < 2.228$, so $0.025 <$ P-value < 0.05: reject H_0

Why test at the boundary?



Largest P-value at the boundary μ = 1: reject there, and we reject all of H_0

Example: the baker's bread

A baker claims his bread rises more than 15 cm, on average. His customers doubt it, so he bakes 10 loaves. Heights are normal. (OpenStax, Example 9.9)

$n = 10$; $\bar{x} = 17$ cm; $\sigma = 0.5$ cm; $\alpha = 0.05$

(a) What are H_0 and H_A? z or t?
(b) What are the standard error and the test statistic?
(c) What is the P-value? What is the decision?

(a) $H_0: \mu \leq 15$ $H_A: \mu > 15$, test at $\mu_0 = 15$; σ known, so z
(b) $SE = 0.5 / \sqrt{10} \approx 0.1581$; $z_{obs} = (17 - 15) / 0.1581 \approx 12.65$
(c) $P(Z \geq 12.65 \mid \mu = 15) \approx 0 < 0.05$: reject H_0; the loaves average more than 15 cm

Practice: is the mean greater than 12?

A normal population has σ = 1. We want to verify a claim that its mean is greater than 12. (OpenStax, Try It 9.9)

$n = 36$; $\bar{x} = 12.5$; $\sigma = 1$; $\alpha = 0.05$

(a) What are H_0 and H_A?
(b) What are the standard error and z_{obs}?
(c) What is the P-value? What is the decision?

(a) $H_0: \mu \le 12$ $H_A: \mu > 12$, test at $\mu_0 = 12$
(b) $SE = 1 / \sqrt{36} \approx 0.1667$; $z_{obs} = (12.5 - 12) / 0.1667 = 3.00$
(c) $P(Z \ge 3.00 \mid \mu = 12) = 1 - 0.9987 = 0.0013 < 0.05$: reject H_0